

CLINICAL ECG REPORT -- CONFIDENTIAL

Report Generated: 2026-06-05 05:08:07 UTC

DeepCardio-RAG v2.0 | AI-Assisted, Physician Verified

PATIENT & ENCOUNTER INFORMATION

Patient Name:	John Doe	MRN:	JHH-MRN-2024-001	Encounter ID:	ENC-2024-0331-007
Date of Birth:	1968-04-15	Age / Sex:	56y / Male	Encounter Date:	2026-06-05 05:08:07
Ward / Room:	Cardiac Care Unit, Floor 4B	Weight / Ht:	78 kg / 175 cm	Weight / Ht:	N/A
Ordering MD:	Dr. Sarah Chen, MD	Interpreting MD:	Dr. Michael Park, MD, PhD	Department:	Division of Cardiology, Johns Hopkins
Clinical Indication:	Chest pain x2h, elevated troponin-I (2.8 ng/mL), rule out STEMI				

TECHNICAL ACQUISITION PARAMETERS

Lead System: Standard 12-Lead | Sampling Rate: 500 Hz | Duration: 10.0s | Bandwidth: 0.05-150 Hz | Notch: 60 Hz | Calibration: 1.0 mV | Device: GE MAC 5500 HD

SECTION 1 -- RHYTHM ANALYSIS

Rhythm:	Sinus Tachycardia with frequent PVCs
Heart Rate:	102 bpm
Regularity:	Regular
HRV:	Normal

SECTION 2 -- INTERVALS & AXES

PR Interval:	164 ms (Normal: 120-200 ms)
QRS Duration:	98 ms (Normal: <120 ms)
QT Interval:	412 ms
QTc Interval:	424 ms (Normal: <=440 ms men / <=460 ms women)
P-wave Axis:	60 deg (Normal: 0-75 deg)
QRS Axis:	-15 deg (Normal: -30 to +90 deg)
T-wave Axis:	35 deg

SECTION 3 -- WAVEFORM MORPHOLOGY

P Waves:	Normal morphology, upright in I, II, aVF
PR Comment:	Normal (120-200 ms)
QRS Complex:	RSR prime in V1-V2 (incomplete RBBB); pathological Q-waves in II, III, aVF
QT/QTc:	Normal QTc (<=440 ms)
ST Segment:	ST elevation 2.5mm V2-V4, 1mm V5; reciprocal depression 1mm in II, III, aVF
T Waves:	Upright in I, II, V4-V6; inverted in aVR (normal)
U Waves:	Not identified

SECTION 4 -- PRIMARY IMPRESSION

ANTERIOR STEMI PATTERN -- CRITICAL: Activate cath lab immediately

ICD-10 Codes: R00.0

RISK CATEGORY: CRITICAL - HIGH RISK: ST elevation meets STEMI criteria (2023 AHA/ACC). Immediate

*** EMERGENT ACTION REQUIRED -- NOTIFY ATTENDING IMMEDIATELY ***

SECTION 5 -- DIFFERENTIAL DIAGNOSIS

- Proximal LAD occlusion (most likely -- Door-to-Balloon <90 min)
- Takotsubo (stress) cardiomyopathy -- echo to differentiate

3. Myocarditis -- less likely given focal ST changes
4. Early repolarisation variant -- excluded by degree of elevation

SECTION 6 -- CLINICAL SIGNIFICANCE & RISK STRATIFICATION

HIGH RISK: ST elevation meets STEMI criteria (2023 AHA/ACC). Immediate revascularisation required. Door-to-Balloon <90 min.

SECTION 7 -- RECOMMENDATIONS

1. IMMEDIATE: Activate cardiac catheterisation lab (Code STEMI)
2. Dual antiplatelet: Aspirin 325mg + Ticagrelor 180mg loading dose
3. Anticoagulation: UFH 60 U/kg bolus (max 4000 U)
4. Serial 12-lead ECG every 30 minutes
5. Serial troponin every 3 hours x3
6. Bedside echocardiography for wall motion assessment
7. Cardiology attending at bedside stat

SECTION 8 -- EVIDENCE-BASED REFERENCES

[1] 2023 AHA/ACC STEMI Guidelines -- doi:10.1161/CIR.0000000000001123

[2] ESC 2022 Acute Coronary Syndromes Guidelines

[3] PLATO Trial -- Ticagrelor vs Clopidogrel in ACS (NEJM 2009)

AI-ASSISTED ANALYSIS METADATA

Model: DeepCardio-RAG v2.0 (ResNet-34 + RAG Attention)

Overall AI Confidence: 96.1% | Inference Time: 0 ms

NLG Quality -- BLEU-4: 0.742 | ROUGE-L: 0.831 | Hallucination Rate: 4.10% (target <5%)

PHYSICIAN VERIFICATION & ELECTRONIC SIGNATURE

Interpreting Physician: Dr. Michael Park, MD, PhD, FACC Physician ID / NPI: JHH-MD-3309

Department: Division of Cardiology, Johns Hopkins Hospital Institution: Johns Hopkins Hospital

This report has been reviewed and electronically signed by the interpreting cardiologist. Electronic signature complies with 21 CFR Part 11 and HIPAA requirements.

Physician Signature Date / Time: 2026-06-05 05:08

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